SOLUTION

SSECTION–I

1. Ans. B

$g(x) = f^{-1}(x) \Rightarrow fog(x) = x$

$\Rightarrow f'(g(x))g'(x) = 1 \Rightarrow g'(x) = \frac{1}{f'(g(x))}$

$\Rightarrow g'(2) = \frac{1}{f'(g(2))} = \frac{1}{f'(a)} = \frac{1+a^2}{a^{10}}$

2. Ans.(C)

By eliminating θ, we get

$y^2 - x^2 = 4 \Rightarrow 2y\frac{dy}{dx} - 2x = 0$

$\Rightarrow y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 1 \Rightarrow y\frac{d^2y}{dx^2} + \frac{x^2}{y^2} = 1$

$y^3\frac{d^2y}{dx^2} = y^2 - x^2 = 4$

3. Ans. (B)

$f(x) = \sqrt{(x-1)+4-4\sqrt{x-1}} + \sqrt{(x-1)+9-6\sqrt{x-1}}$

$= |\sqrt{x-1}-2| + |\sqrt{x-1}-3|$

at x=1.5

$f(x) = (2-\sqrt{x-1}) + (3-\sqrt{x-1})$

$= 5 - 2\sqrt{(x-1)}$

$f'(x) = \frac{-2}{2\sqrt{x-1}}$

$f'(1.5) = -\frac{1}{\sqrt{\frac{3}{2}-1}} = -\sqrt{2}$

4. Ans. (A)

Slope of normal = 3

$\Rightarrow$ slope of tangent $f'(0) = -\frac{1}{3}$

$L = \lim_{x\to1}\frac{(x-1)^2}{f(x-1)^2 - 5f\{4(x-1)^2\} + 4f\{7(x-1)^2\}}$

$\left(\frac{0}{0}\text{ form}\right)$

$= \lim_{x\to1}\frac{2(x-1)}{2(x-1)f'(x-1) - 40f'\{4(x-1)^2\}(x-1) + 56f'\{7(x-1)^2\}(x-1)}$

$= \lim_{x\to1}\frac{2}{2f'(x-1) - 40f'\{4(x-1)^2\} + 56f'\{7(x-1)^2\}}$

$= \frac{1}{f'(0) - 20f'(0) + 28f'(0)} = \frac{1}{-\frac{1}{3} + \frac{20}{3} - \frac{28}{3}} = -\frac{1}{3}$

5. Ans. D

$\because$ Function is continuous at x = 0 then

$f(0) = \lim_{x\to0}(x+1)^{\cot x}$ (1^∞ for m)

$= e^{\lim_{x\to0}\cot x(x+1-1)}$

$= e^{\lim_{x\to0}\frac{x}{\tan x}} = e$

6. Ans. (A)

Let x = 1 + t

$\lim_{t\to0}\frac{\ell n(\sin^{-1}(\ell n(1+t))}{1/t} \to \frac{\infty}{\infty}$

$\lim_{t\to0}\frac{1}{\sin^{-1}(\ell n(1+t))}\frac{1}{\sqrt{1-\ell n^2(1+t)}}\cdot\frac{1}{(1+t)\left(-\frac{1}{t^2}\right)}$

$= -\lim_{t\to0}\frac{t^2}{\sin^{-1}\ell n(1+t)} = -\lim_{t\to0}\frac{t^2}{\sin^{-1}\ell n(1+t)}\frac{\ell n(1+t)}{\ell n(1+t)}$

$= -\lim_{t\to0} t.\frac{t}{\ell n(1+t)} = 0$

7. Ans. B

$\because g(x) = g(-x) \Rightarrow g'(x) = -g'(-x)$

$\Rightarrow g'(0) = 0$

$f'(0^+) = \lim_{h\to0}\frac{g(0+h)\cos\left(\frac{1}{0+h}\right) - f(0)}{h}$

$= \lim_{h\to0}\frac{g(h)\cos\left(\frac{1}{h}\right)}{h} = \lim_{h\to0}\frac{g(h)-g(0)}{h}\cos\frac{1}{h}$

($\because g(0) = 0$)

= g'(0)x a finite value

$f'(0^-) = \lim_{h\to0} = \frac{g(0-h)\cos\left(\frac{1}{0-h}\right) - f(0)}{-h}$

$= \lim_{h\to0}\frac{g(-h)\cos\left(\frac{-1}{h}\right)}{-h} = 0$ (similarly)

$\therefore f'(0) = 0$

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8. Ans. (C)

$$\text{Numerator} = \lim_{x\to\infty}\cos^{-1}\left(\frac{\sqrt{x+\sqrt{x}}-\sqrt{x}}{\sqrt{x+\sqrt{x+\sqrt{x}}}+\sqrt{x+\sqrt{x}}}\right)$$

$$= \lim_{x\to\infty}\cos^{-1}\left(\frac{\sqrt{x}}{\left(\sqrt{x+\sqrt{x+\sqrt{x}}}+\sqrt{x+\sqrt{x}}\right).\left(\sqrt{x+\sqrt{x}}+\sqrt{x}\right)}\right)$$

$$= \cos^{-1}0 = \frac{\pi}{2}$$

$$\text{Denominator} = \lim_{x\to\infty}\sec^{-1}\left(-\ell n\left(\frac{x^2+2}{x^2+1}\right)^{x^2}\right) =$$

$$\sec^{-1}\left(-\ell n e^{\lim\limits_{x\to\infty}\frac{x^2}{x^2+1}}\right) = \sec^{-1}(-1) = \pi$$

$$\therefore \text{ Required limit} = \frac{\pi/2}{\pi} = \frac{1}{2}$$

9. Ans. (D)

$$\lim_{n\to\infty}\frac{g\left(2009+\frac{1}{n}\right)-g(2009)}{\frac{1}{n}} = 27 \Rightarrow g'(2009)$$

$$= 27$$

$$\lim_{\alpha\to\frac{\pi}{2}} g(2009+\cos\alpha) = 3 \quad \Rightarrow \quad g(2009) = 3$$

$$\lim_{t\to0^+}\left(\frac{g(2009)}{g(2009+t)}\right)^{1/t} \text{ is } 1^\infty \text{ form}$$

$$e^{\lim\limits_{t\to0}\frac{1}{t}\left(\frac{g(2009)-g(2009+t)}{g(2009+t)}\right)} \Rightarrow$$

$$e^{-\lim\limits_{t\to0}\left(\frac{g(2009+t)-g(2009)}{t}\right)\frac{1}{g(2009+t)}} = e^{\frac{-g'(2009)}{g(2009)}} \Rightarrow e^{-9}$$

10. Ans.(D)

function is always continuous at every point in its domain.

11. Ans. A,C

$f^{-1}(x) = g(x) \quad \Rightarrow \quad x = f(g(x))$

Differentiating both sides,

$$1 = f'(g(x))\,g'(x) \quad \Rightarrow \quad g'(x) = \frac{1}{f'(g(x))}$$

Now $f'(x) = 2x + 3$

$$\text{So } g'(x) = \frac{1}{2g(x)+3} \quad \Rightarrow \quad g'(1) = \frac{1}{2(g(1))+3}$$

$\because f(x) = x^2 + 3x - 3 \Rightarrow f(1) = 1 \Rightarrow g(1) = 1$

$\Rightarrow \quad g(1) = 1$

$$\Rightarrow \quad g'(1) = \frac{1}{5}$$

12. Ans. (A,B,C)

(A) for $a = 0$

$$L = \lim_{x\to\infty}\left(\frac{2}{\sqrt{4+x^2}}\right)^{\frac{1}{x^2+1}} \qquad (0^0 \text{ from})$$

$$\log L = \lim_{x\to\infty}\frac{1}{x^2+1}\log\left(\frac{2}{\sqrt{4+x^2}}\right)$$

$$\log L = \lim_{x\to\infty}\frac{\log 2 - \frac{1}{2}\log(4+x^2)}{x^2+1} \quad \left(\frac{0}{0}\text{form}\right)$$

$$= \lim_{x\to\infty}\frac{-\frac{x}{4+x^2}}{2x} \text{ (applying L'Hopital's rule)}$$

$$= 0$$

$$\Rightarrow \quad L = e^0 = 1$$

(B) For $a = 1$

$$L = \lim_{x\to\infty}\left(\frac{x+2}{\sqrt{4+x^2}}\right)^{\frac{x^3+1}{x^2+1}} \qquad (1^\infty \text{ form})$$

$$L = e^{\lim\limits_{x\to\infty}\left(\frac{x+2-\sqrt{4+x^2}}{\sqrt{4+x^2}}\right)\frac{x^3+1}{x^2+1}}$$

$$L = e^{\lim\limits_{x\to\infty}\left(\frac{4x}{\sqrt{4+x^2}\left(x+2+\sqrt{4+x^2}\right)}\right)\frac{x^3+1}{x^2+1}}$$

$$L = e^{\lim\limits_{x\to\infty}\left(\frac{4}{\sqrt{\frac{4}{x^2}+1}\left(1+\frac{2}{x}+\sqrt{\frac{4}{x^2}+1}\right)}\right)\left(\frac{1+\frac{1}{x^3}}{1+\frac{1}{x^2}}\right)}$$

$$L = e^{4/2} = e^2$$

(C) For $0 < a < 1$ $\qquad L = \lim\limits_{x\to\infty}(a)^\infty = 0$

13. Ans. B,D

Since $\sin^{-1} x$ and $\cos\frac{1}{x}$ are continuous & differentiable in $x \in [-1, 1] - \{0\}$

Now at $x = 0$

$$f'(0^-) = \lim_{h\to0}\frac{(\sin^{-1}(0-h))^2\cos\left(\frac{-1}{h}\right)-0}{-h} = 0$$

$$f'(0^+) = \lim_{h\to0}\frac{(\sin^{-1}h)^2\cos\left(\frac{1}{h}\right)-0}{h} = 0$$

Hence LHD = RHD

so f(x) is continuous & differentiable every where in $-1 \le x \le 1, x \ne 0$

14. Ans. B,C

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Paragraph for Question 15 to 16

15. Ans. (B)

$F'(x) = -1$ (constant function) in $(-4,-1)$

$\Rightarrow F(x) = -x + c$ i.e. linear function

16. Ans. (C)

From graph, it is clear that $F'(x)$ is not defined at $x = -4,-1$ and 4

SECTION–III

1. Ans. 4

$f'(x) = 3x^2 + 2xf'(1) + f''(2)$ .....(i)

$f''(x) = 6x + 2f'(1)$ .....(ii)

Now put $x = 1$ in(i)

$f'(1) = 3 + 2f'(1) + f''(2)$

$f'(1) + f''(2) + 3 = 0$ .....(iii)

and $x = 2$ in (ii)

$f''(2) = 12 + 2f'(1)$

$2f'(1) - f''(2) + 12 = 0$ .....(iv)

by (iii) & (iv)

$f'(1) = -5,\ f''(2) = 2$

also $f'''(x) = 6 \Rightarrow f'''(4) = 6$ .....(v)

$\Rightarrow f(x) = x^3 - 5x^2 + 2x + 6$

$f(1) = 1 - 5 + 2 + 6 = 4$

2. Ans. 0

$y(x) = \sin(x - \sin x)$

$\Rightarrow y'(x) = \cos(x - \sin x).(1 - \cos x)$

$\Rightarrow y''(x) = (1 - \cos x)[-\sin(x - \sin x)(1 - \cos x)] + \cos(x - \sin x).\sin x$

put $x = \frac{\pi}{2}$

$$\Rightarrow y''\left(\frac{\pi}{2}\right) = -\sin\left(\frac{\pi}{2} - 1\right) + \cos\left(\frac{\pi}{2} - 1\right)$$

$$= -\cos 1 + \sin 1$$

$$\Rightarrow y''\left(\frac{\pi}{2}\right) + \cos 1 = \sin 1$$

$$\Rightarrow \left[y''\left(\frac{\pi}{2}\right) + \cos 1\right] = [\sin 1] = 0$$

3. Ans. 2

$$b = \lim_{x\to 1} \frac{(7 + x^3)^{1/a} - (3 + x^2)^{1/2}}{(x - 1)}$$

if limit exist, then $8^{1/a} = 4^{1/2}$

$$b = \lim_{x\to 1} \frac{(7 + x^3)^{1/3} - (3 + x^2)^{1/2}}{(x - 1)} = -\frac{1}{4}$$

$$b = \lim_{x\to 1} \frac{\left(8 + x^3 - 1\right)^{\frac{1}{3}} - \left(4 + x^2 - 1\right)^{\frac{1}{2}}}{x - 1}$$

$$= \lim_{x\to 1} \frac{2\left[1 + \frac{x^3 - 1}{8}\right]^{\frac{1}{3}} - 2\left[1 + \frac{x^2 - 1}{4}\right]^{\frac{1}{2}}}{x - 1}$$

$$= 2\lim_{x\to 1} \frac{\left[1 + \frac{1}{3}.\left(\frac{x^3 - 1}{8}\right) + \ldots\right] - \left[1 + \frac{1}{2}\left(\frac{x^2 - 1}{4}\right) + \ldots\right]}{x - 1}$$

$$= 2\left(\frac{1}{8} - \frac{1}{4}\right) = -\frac{1}{4}$$

$\Rightarrow a + 4b = 2$

4. Ans. 3

$$f(x) = \sin\left([x]\frac{\pi}{2}\right).\text{sgn}(x - 1)(x - 2)$$

$$= \begin{cases} \sin\left([x]\frac{\pi}{2}\right) & x < 1 \text{ or } x > 2 \\ -\sin\left([x]\frac{\pi}{2}\right) & 1 < x < 2 \\ 0 & x = 1, 2 \end{cases}$$

$$= \begin{cases} 0 & x \in [0, 1) \\ 0 & x = 1 \\ -1 & x \in (1,2) \\ 0 & x = 2 \\ 0 & x \in (2,3) \\ -1 & x = 3 \end{cases} = \begin{cases} 0 & 0 \le x \le 1 \\ -1 & 1 < x < 2 \\ 0 & 2 \le x < 3 \\ -1 & x = 3 \end{cases}$$

$\Rightarrow$ function is discondinuous at $x = 1, 2, 3$

SECTION–IV

1. Ans. (A)→(R); (B)→(Q,S); (C)→(R); (D)→(T)

$$f(x) = \begin{cases} 1, & x = 0 \\ 0, & 0 < x < \frac{\pi}{2} \\ 1, & x = \frac{\pi}{2} \\ -1, & \frac{\pi}{2} < x \le \pi \\ 0, & \pi < x < \frac{3\pi}{2} \\ -1, & \frac{3\pi}{2} \le x \le 2\pi \end{cases}$$

Number of points where $f(x)$ is discontinuous & nonderivable = 4

at $x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$

$\lim_{x\to a} f(x)$ does not exist

for $x = \frac{\pi}{2}, \pi, \frac{3\pi}{2}$

$f(x) = -1$ (Minimum) for $x \in \left(\frac{\pi}{2}, \pi\right] \cup \left[\frac{3\pi}{2}, 2\pi\right]$

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